

Design Report BGR

Maehashi

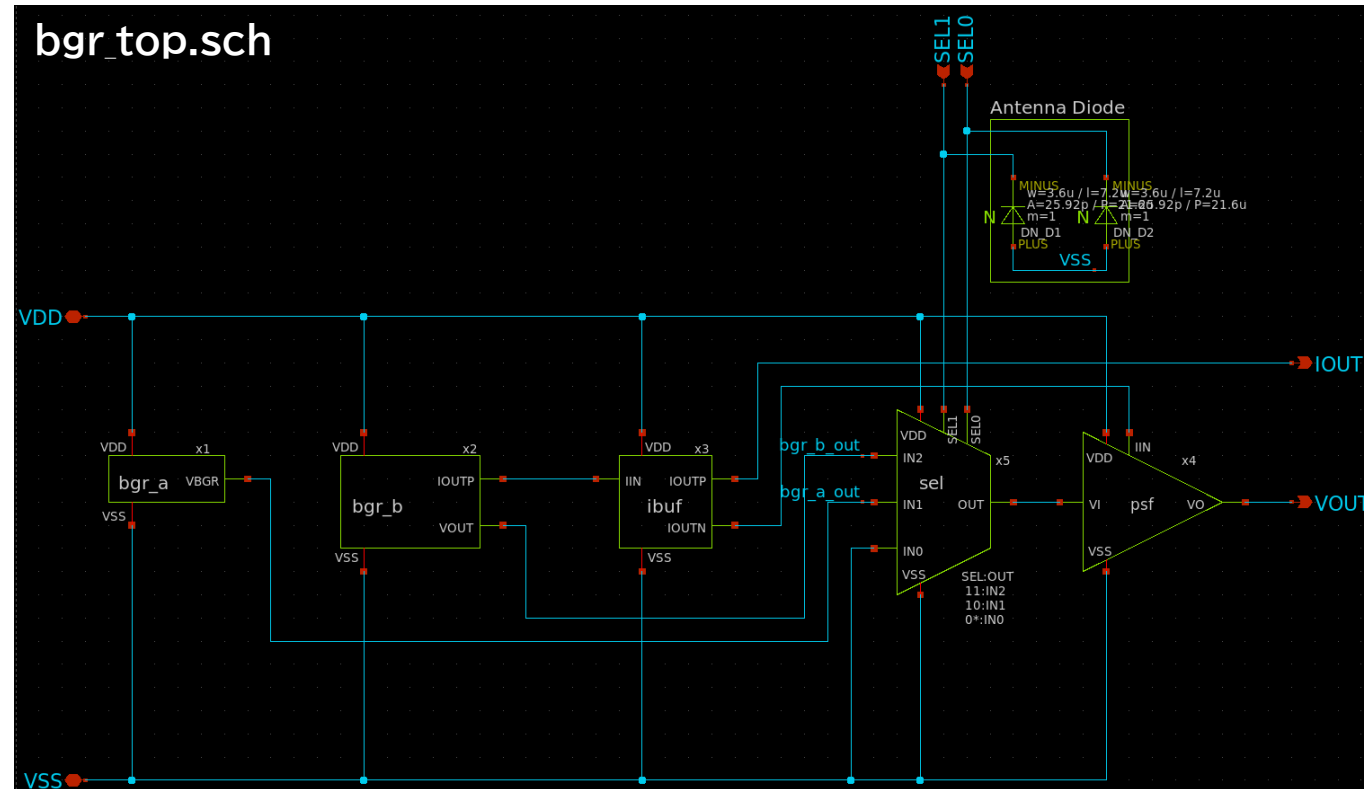


Overview

- Process TR10 (OpenSUSI TR-1um)
- Tools Circuit Design: Xschem, Layout Design: Klayout
- PVT Variation
 - VDD $5.0V \pm 0.5V$
 - Temp. $0^{\circ}C \sim 80^{\circ}C$
 - Process MOS Vth $\pm 0.1V$, Poly-Res. $\pm 15\%$

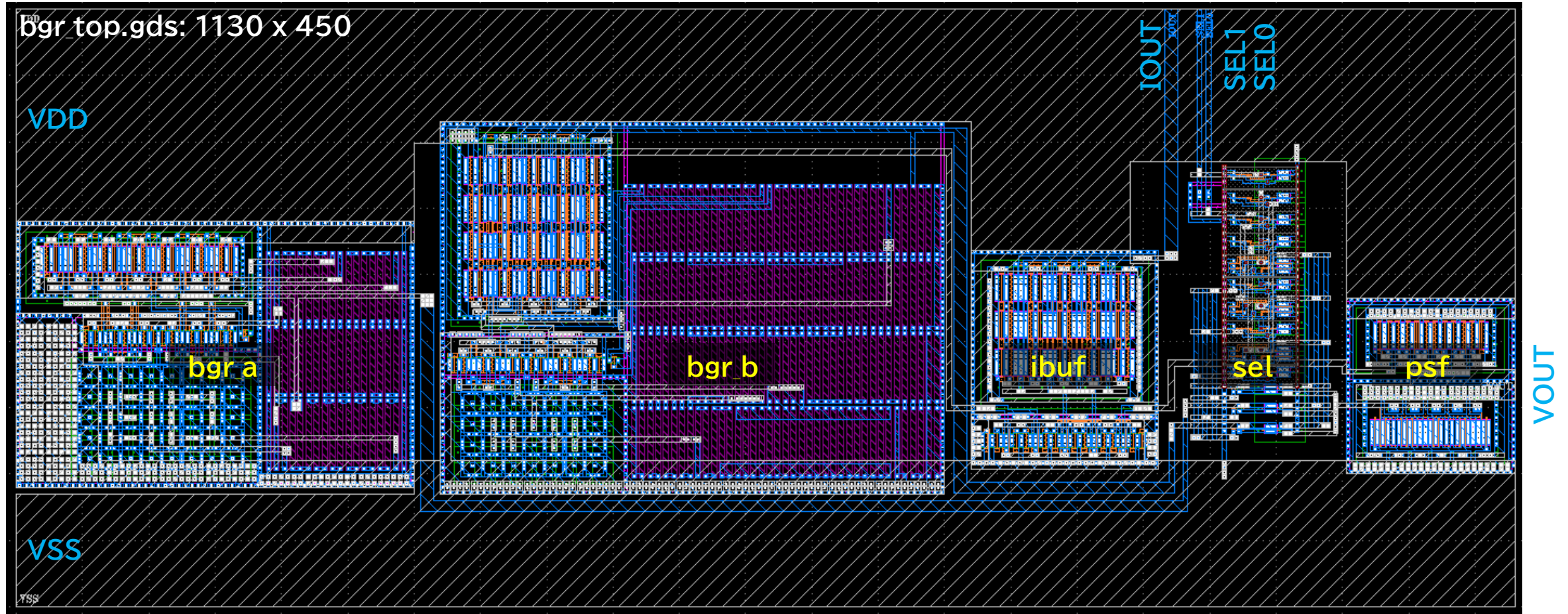
Result		bgr_a			bgr_b		
		0°C	40°C	80°C	0°C	40°C	80°C
Vout	SS/4.5V	1.0876 V	1.0868 V	1.0866 V	2.4643 V	2.4596 V	2.4559 V
	TT/5.0V	1.0948 V	1.0951 V	1.0961 V	2.4796 V	2.4773 V	2.4762 V
	FF/5.5V	1.1129 V	1.1120 V	1.1117 V	2.4937 V	2.4938 V	2.4955 V
Iout	SS/4.5V	-	-	-	38.124 uA	38.050 uA	37.992 uA
	TT/5.0V	-	-	-	45.955 uA	45.912 uA	45.892 uA
	FF/5.5V	-	-	-	54.276 uA	54.278 uA	54.315 uA

Block Diagram



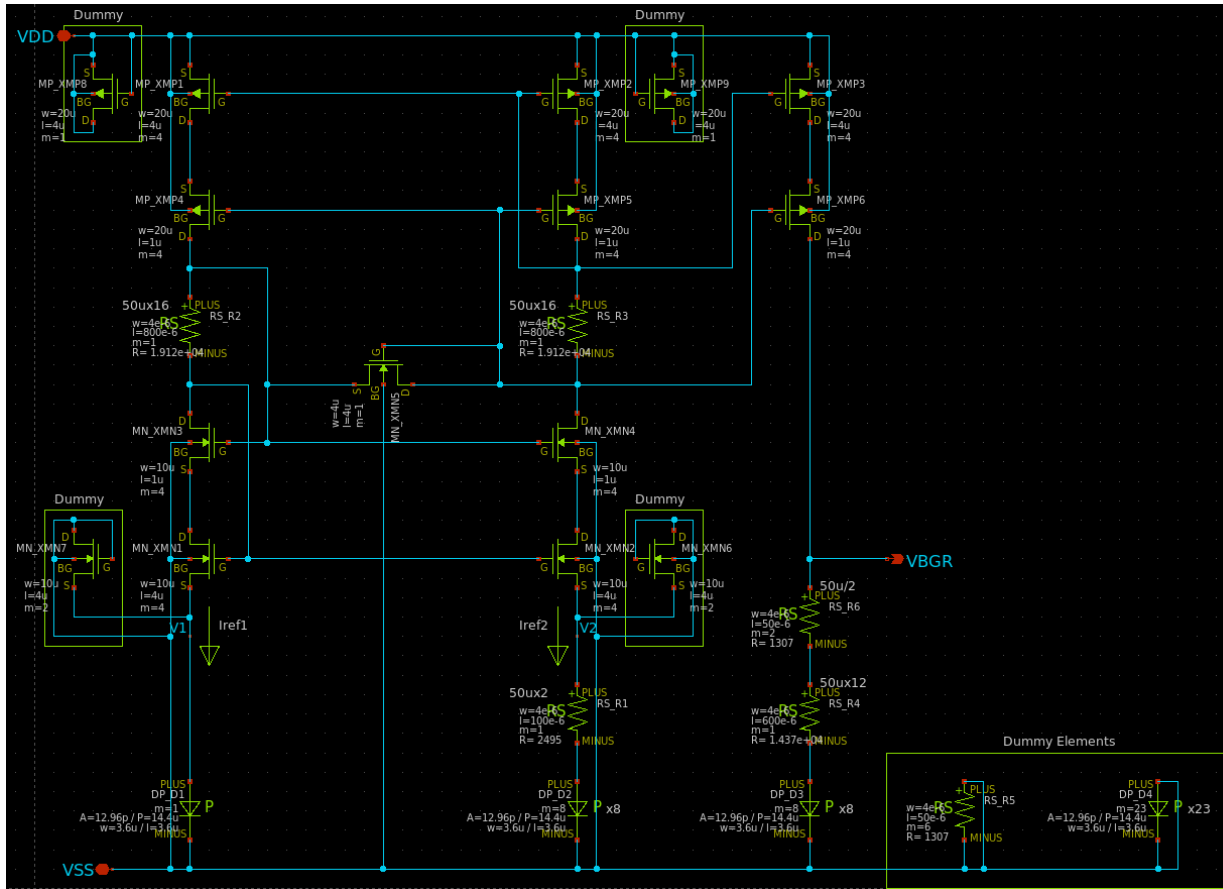
- bgr_a TEG 1
- bgr_b TEG 2
- ibuf for monitoring bgr_b IOUPT
- sel to select output voltage: VBGR(bgr_a)/VOUT(bgr_b)/GND
- psf for monitoring selected voltage

Top Layout



bgr_a

- Schematic (incl. layout-placed dummies)



Architecture^[*]

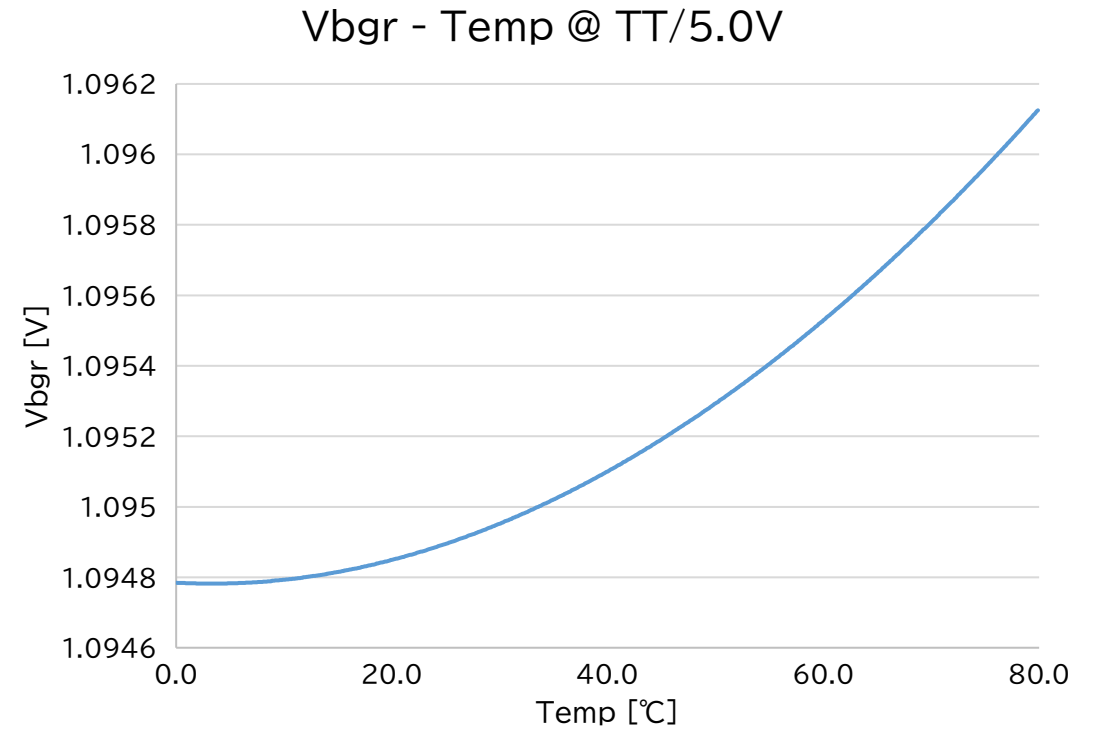
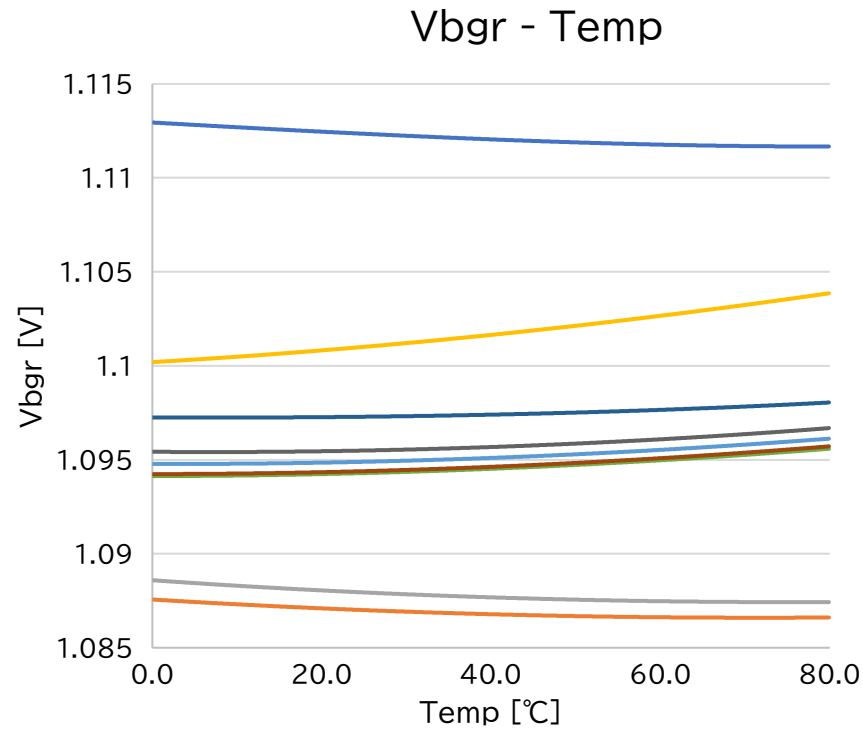
- Iptat Gen. Circuit + Vctat Gen. Diode
- Self-biased Cascode
- Startup Transistor

[*] B. Razavi, "Design of Analog CMOS Integrated Circuit"

bgr_a

Teshbench: tb_bgr_a.sch

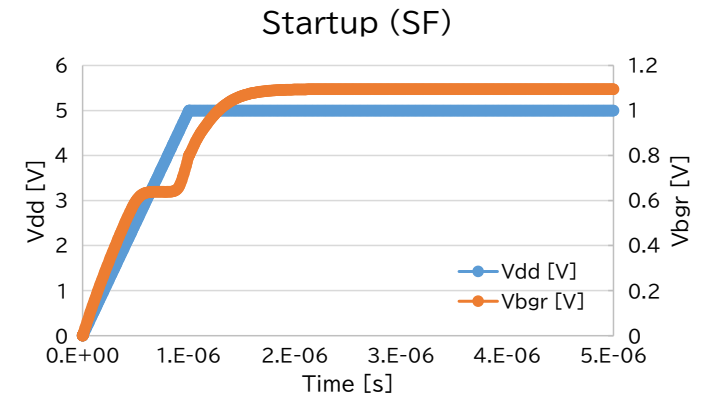
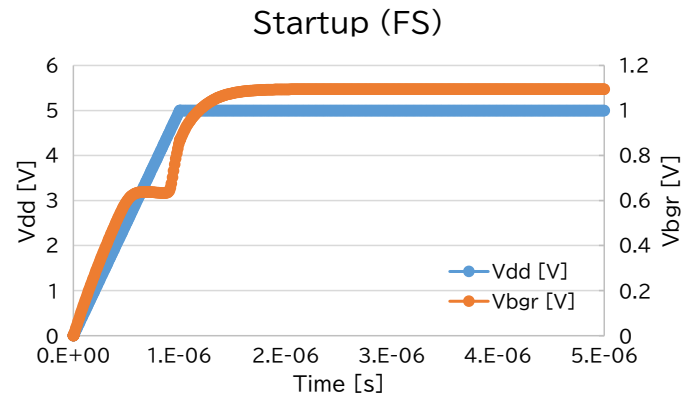
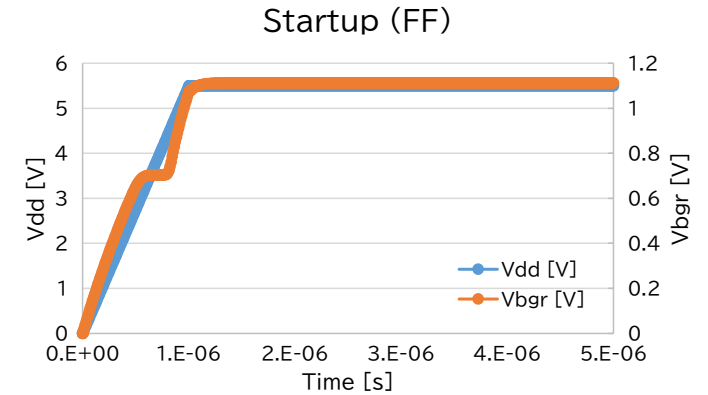
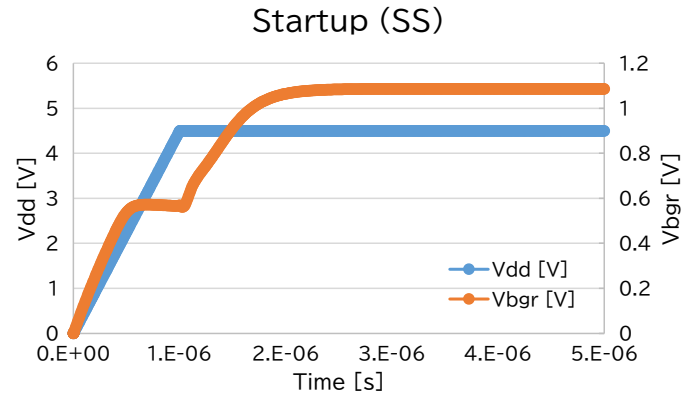
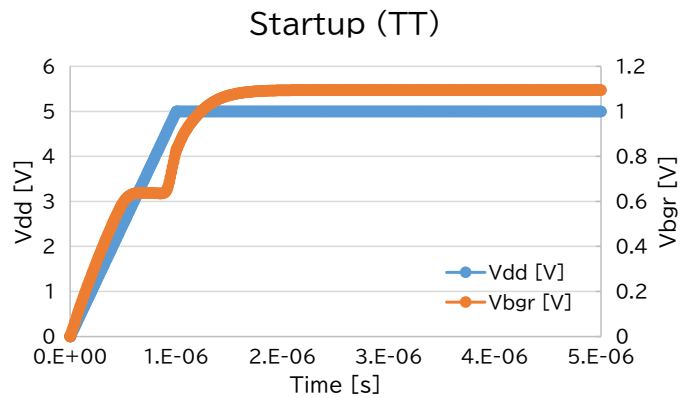
- DC Analysis



bgr_a

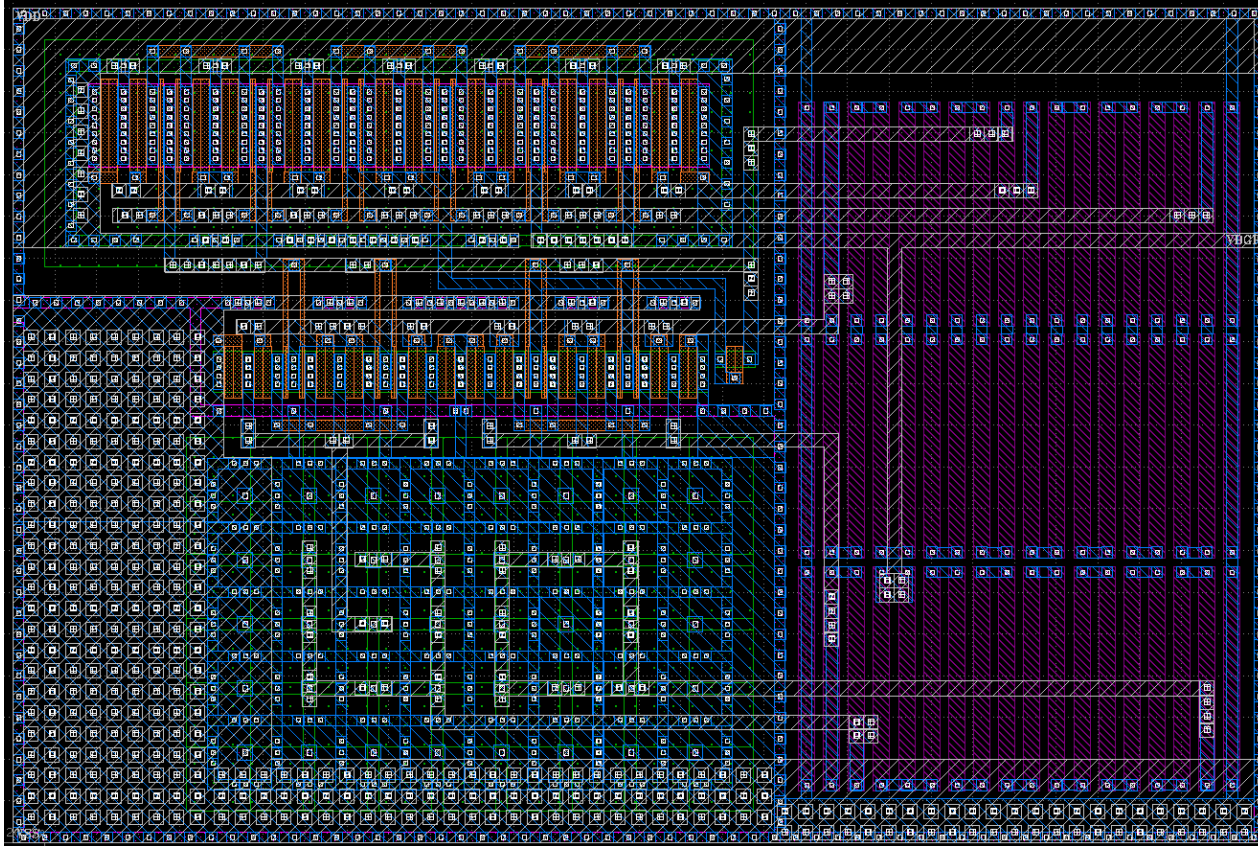
Teshbench: tb_bgr_a.sch

- Tran. Analysis (Startup)



bgr_a

- Layout

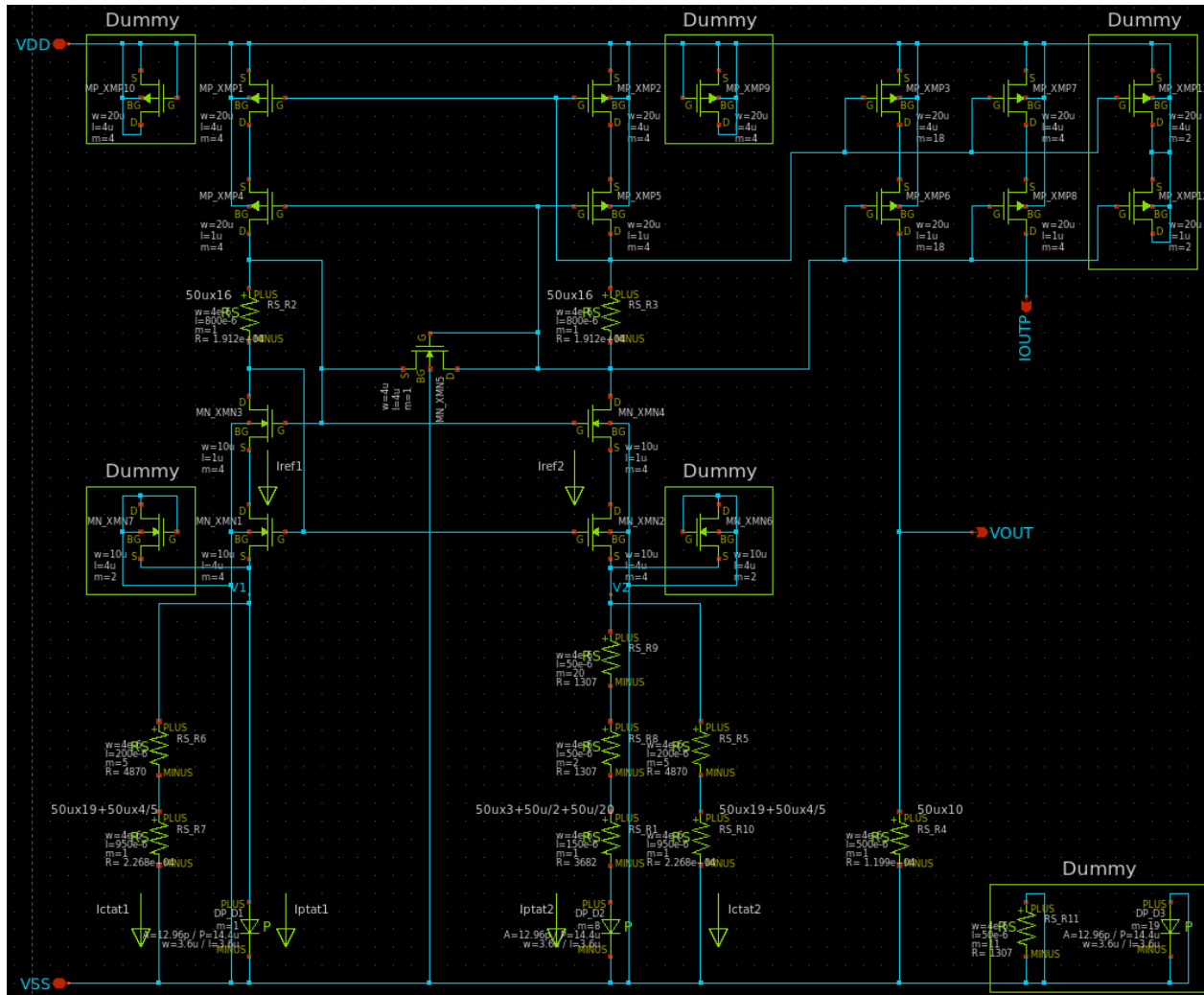


Features

- 300um x 200um Area usage
- Employing poly resistor for stability against temperature variation

bgr_b

- Schematic (incl. layout-placed dummies)



Architecture[*]

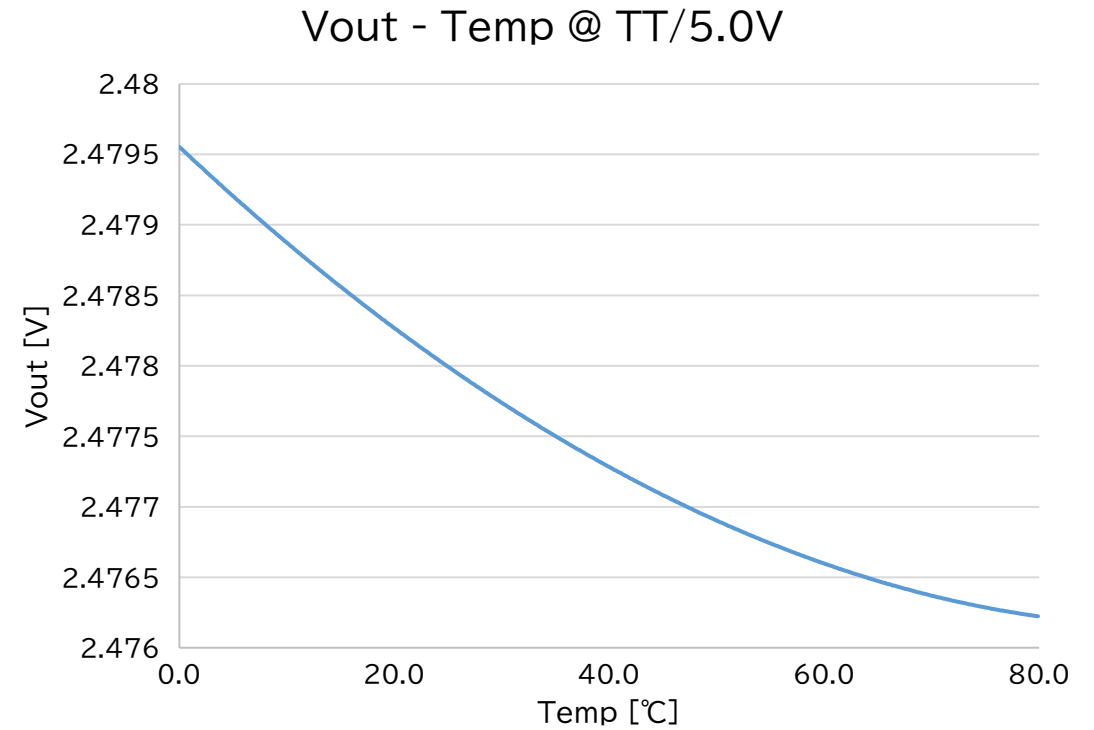
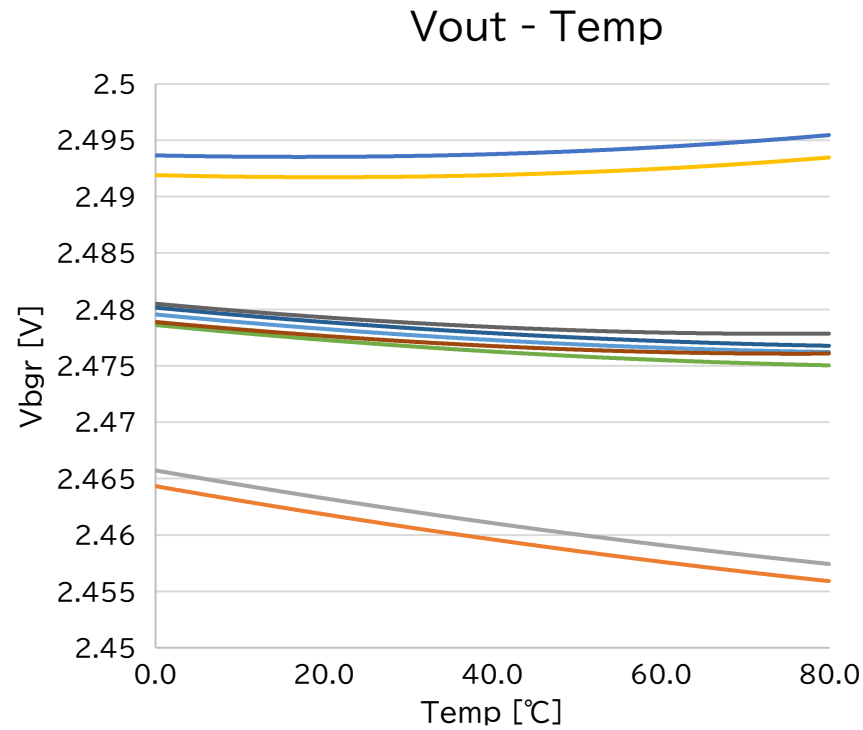
- Iconst Gen. Circuit + Resistor
- Self-biased Cascode
- Startup Transistor

[*] B. Razavi, "Design of Analog CMOS Integrated Circuit"

bgr_b

Teshbench: tb_bgr_b.sch

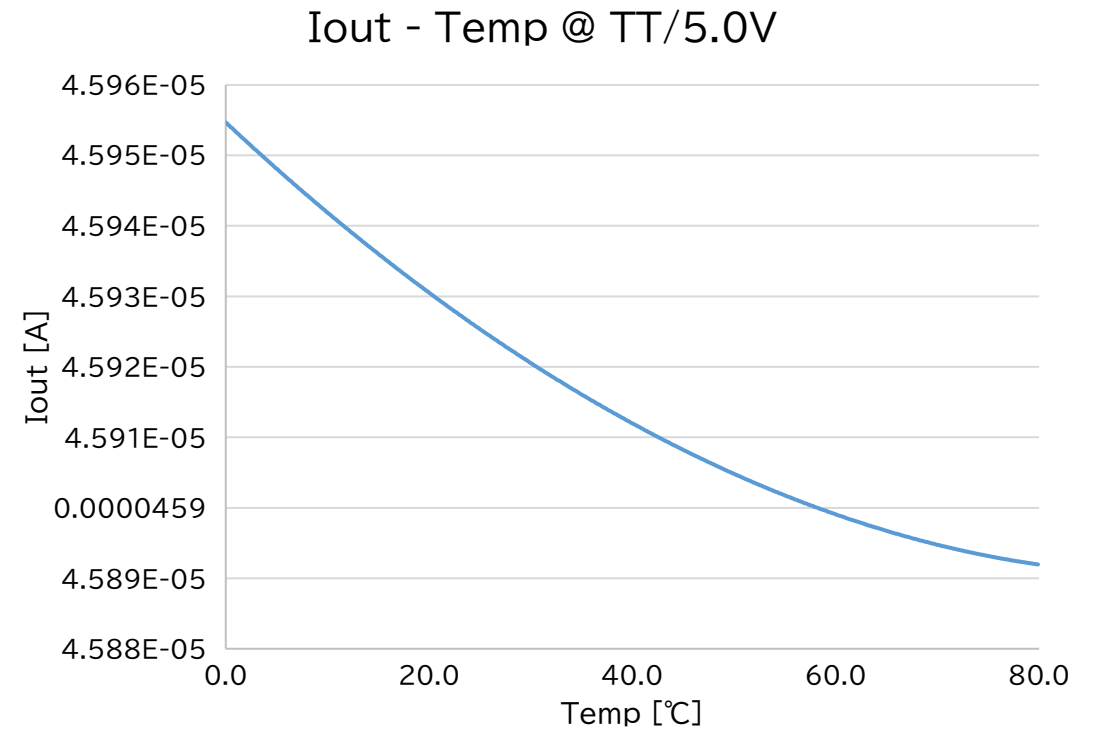
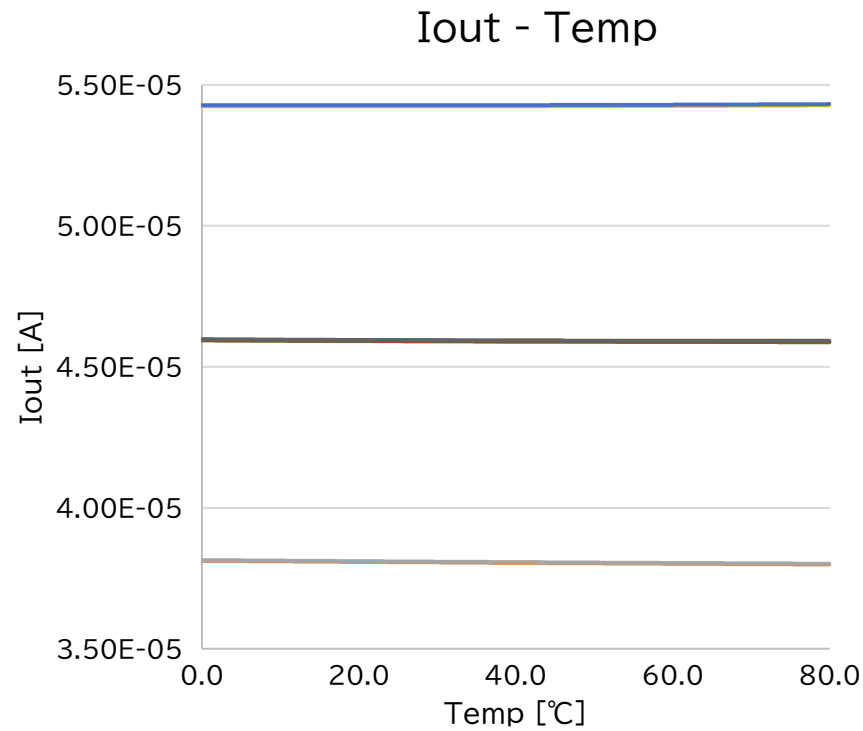
- DC Analysis @ Vout



bgr_b

Teshbench: tb_bgr_b.sch

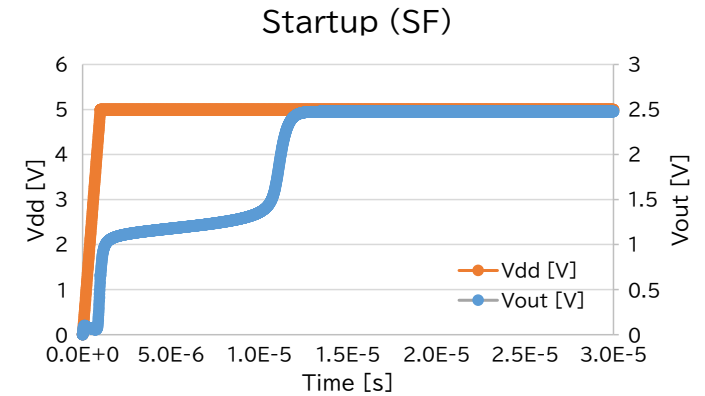
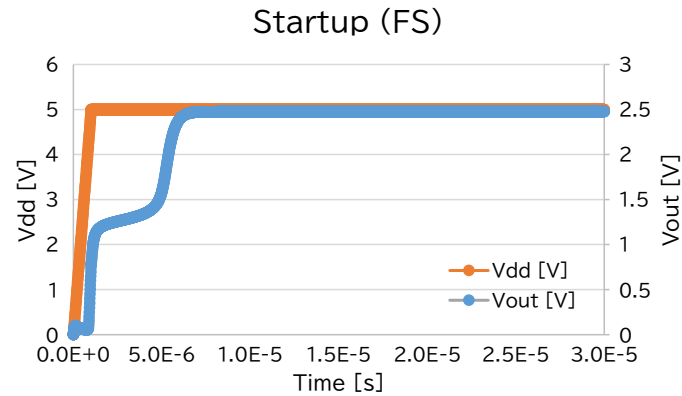
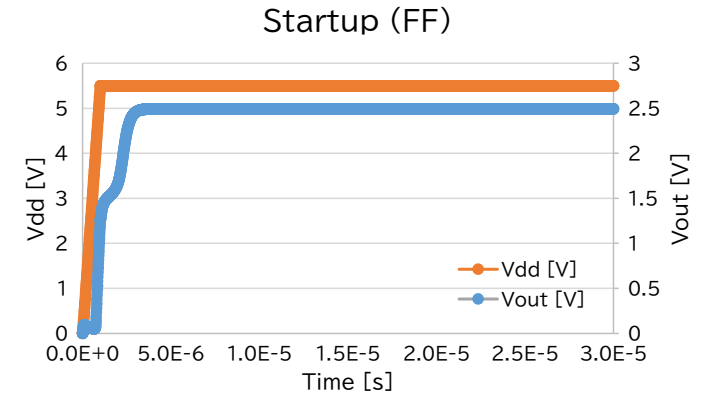
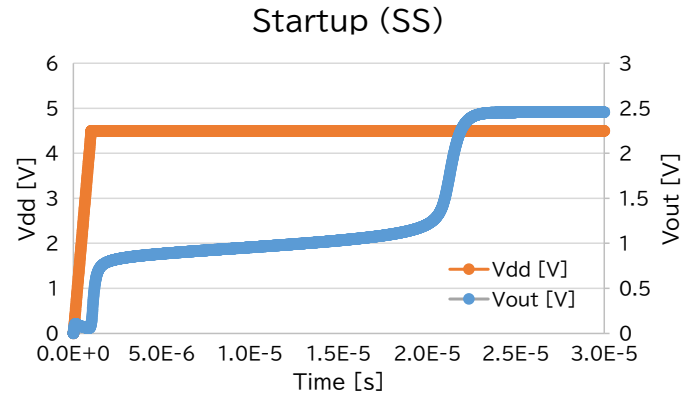
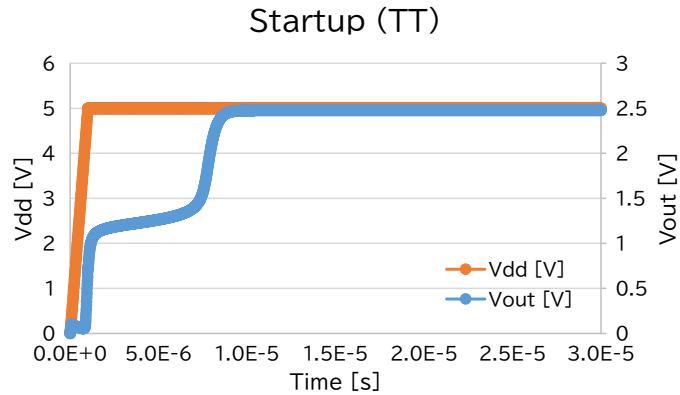
- DC Analysis @ Iout



bgr_b

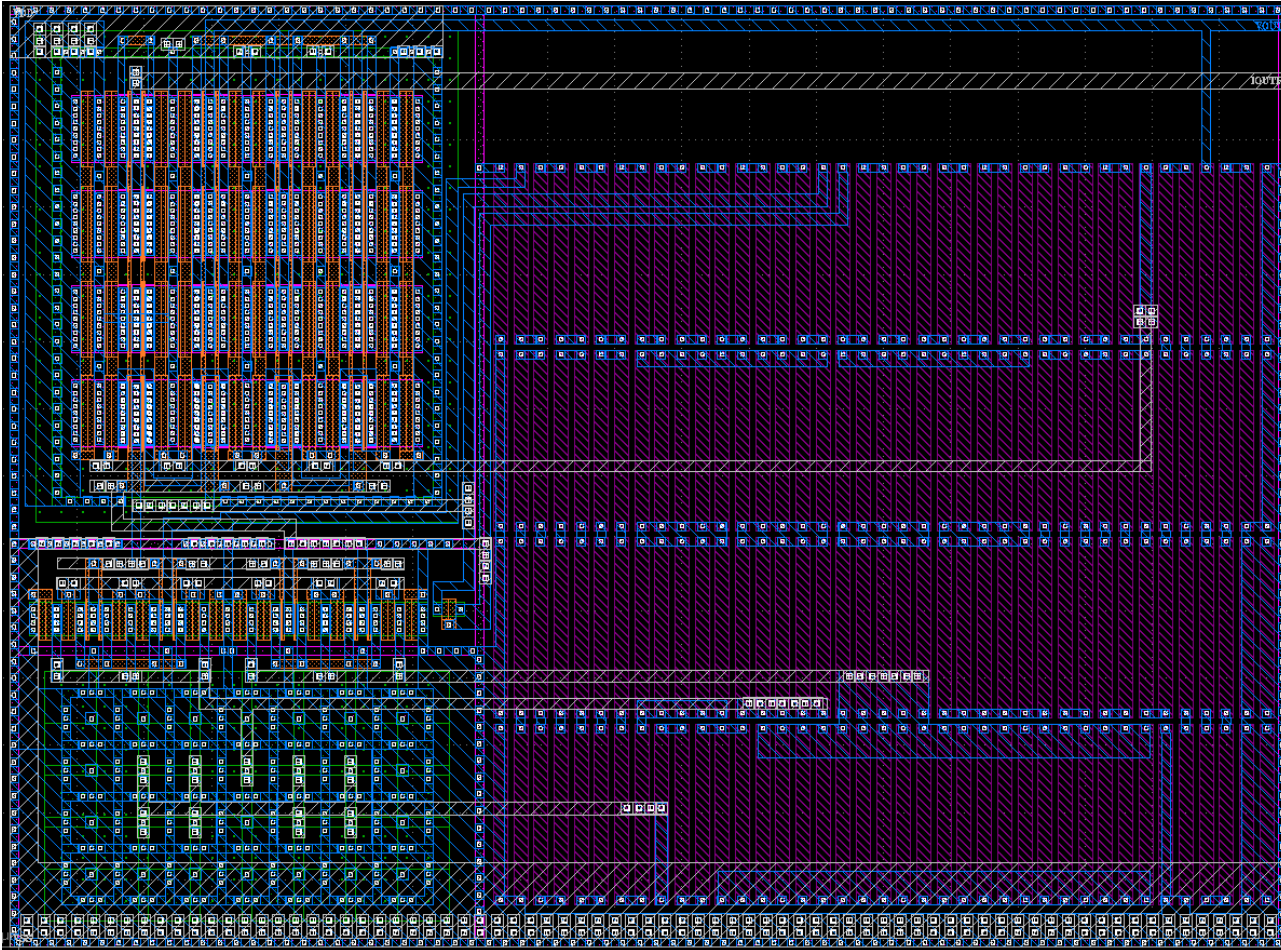
Teshbench: tb_bgr_b.sch

- Tran. Analysis (Startup)



bgr_b

- Layout

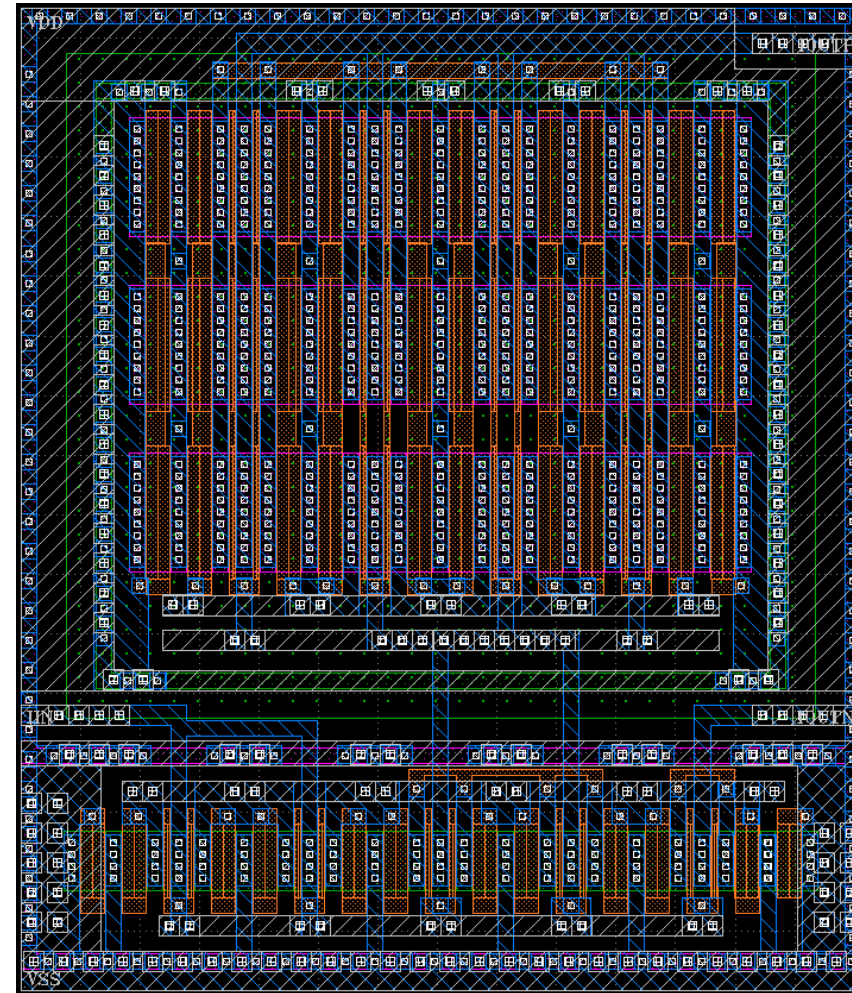
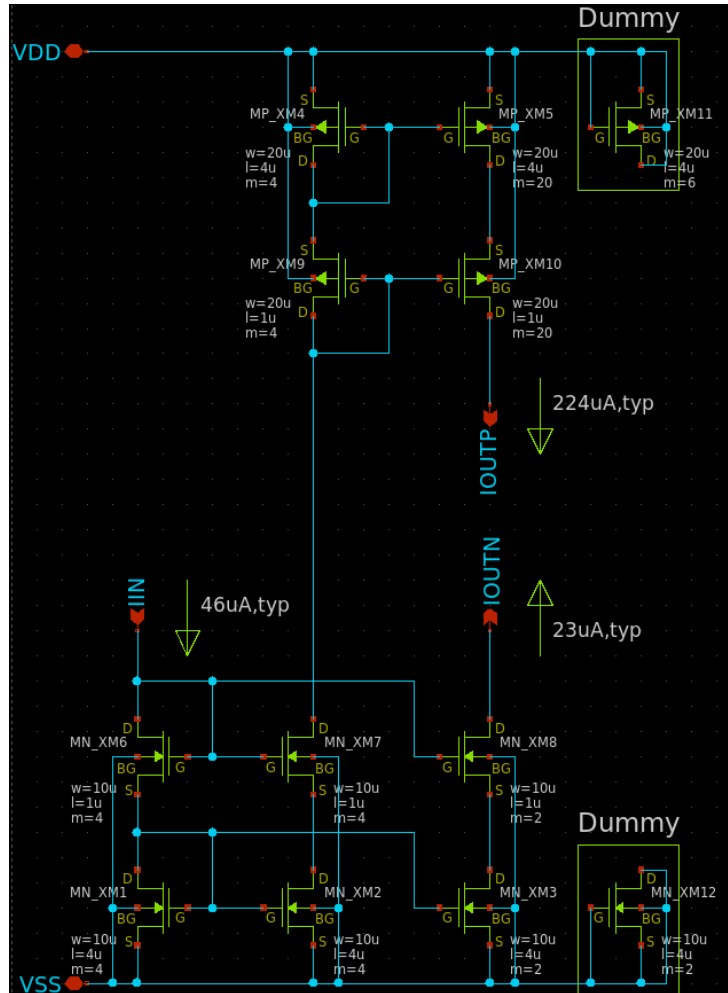


Features

- 380um x 280um Area usage
- Employing poly resistor for stability against temperature variation

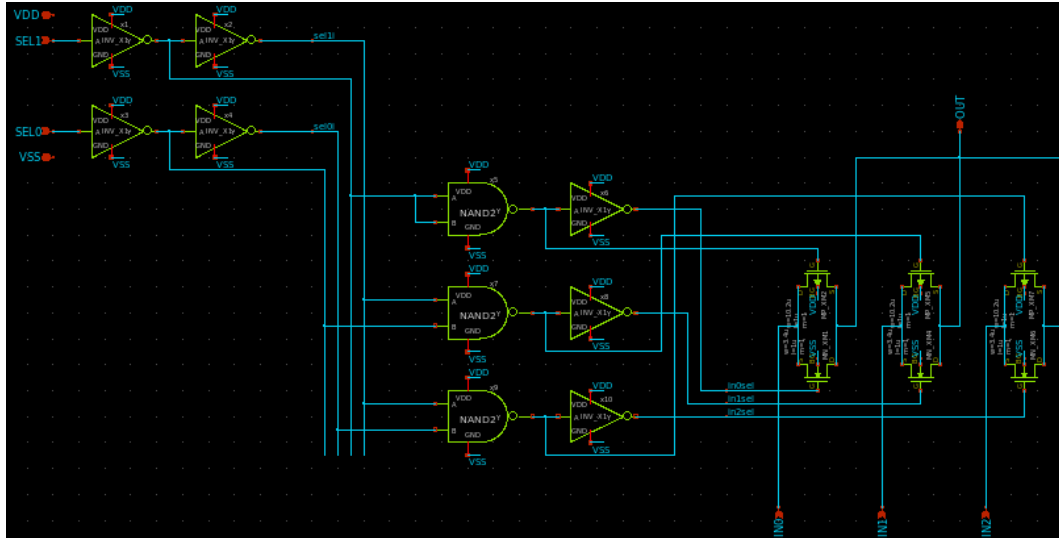
ibuf

- Schematic (incl. layout-placed dummies)
- Layout

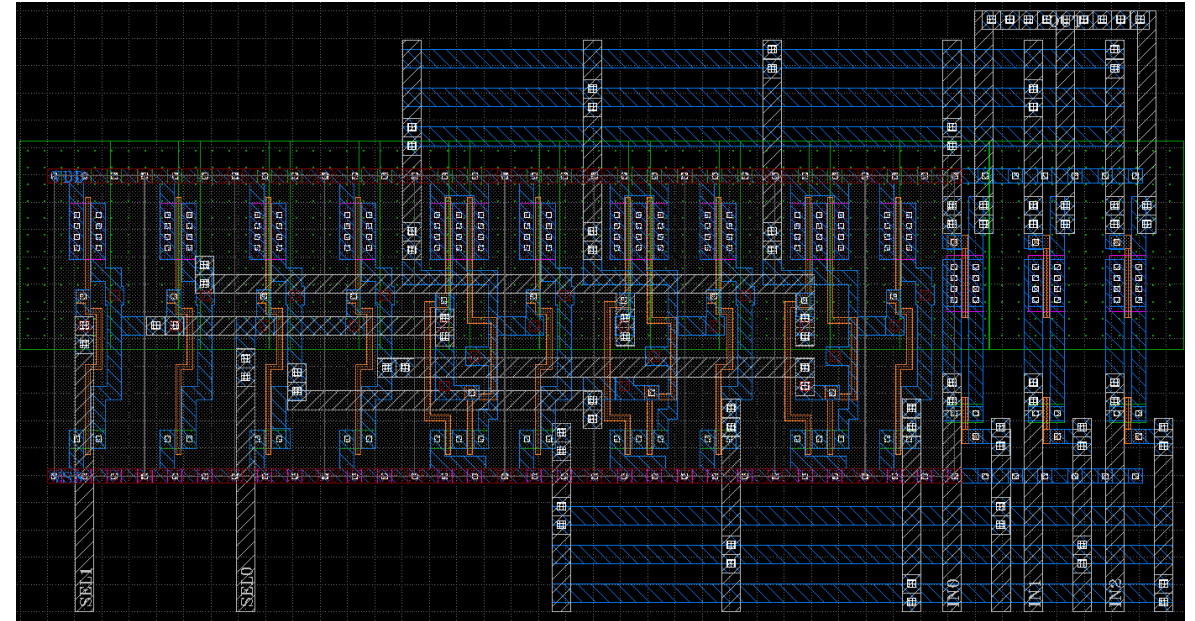


sel

- Schematic (incl. layout-placed dummies)

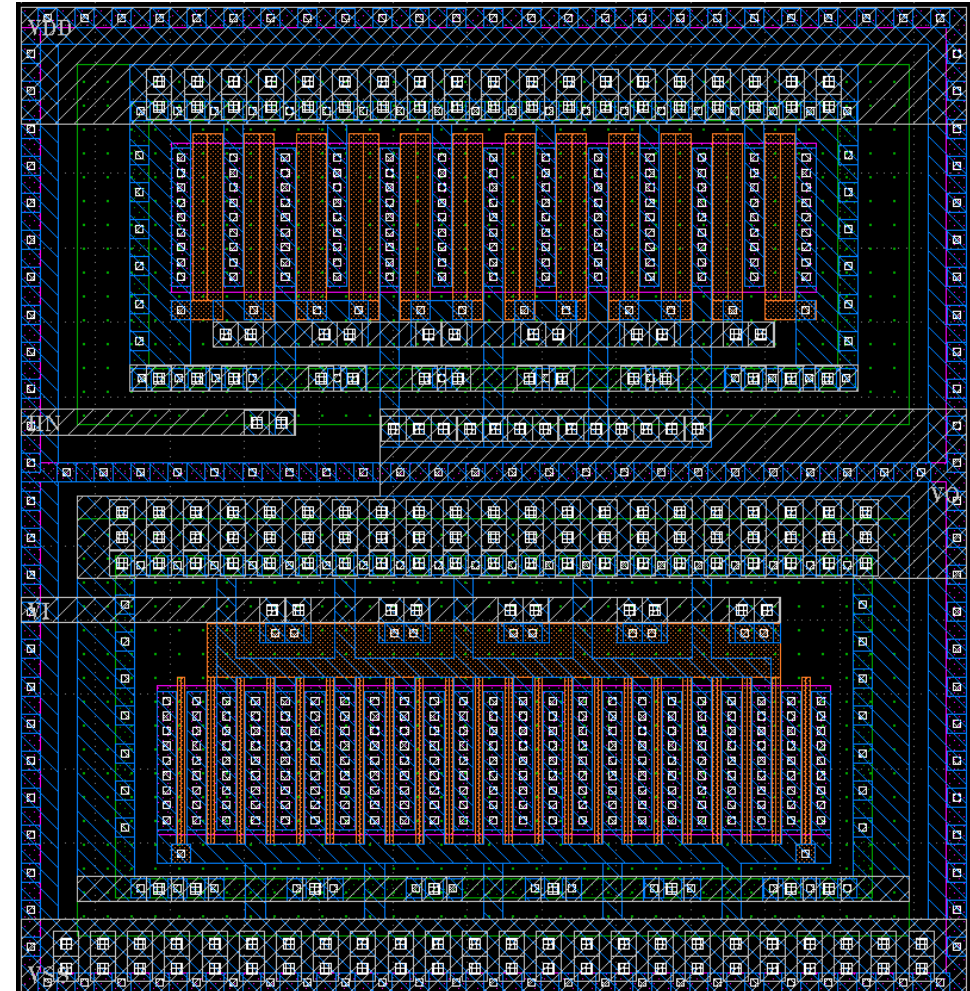
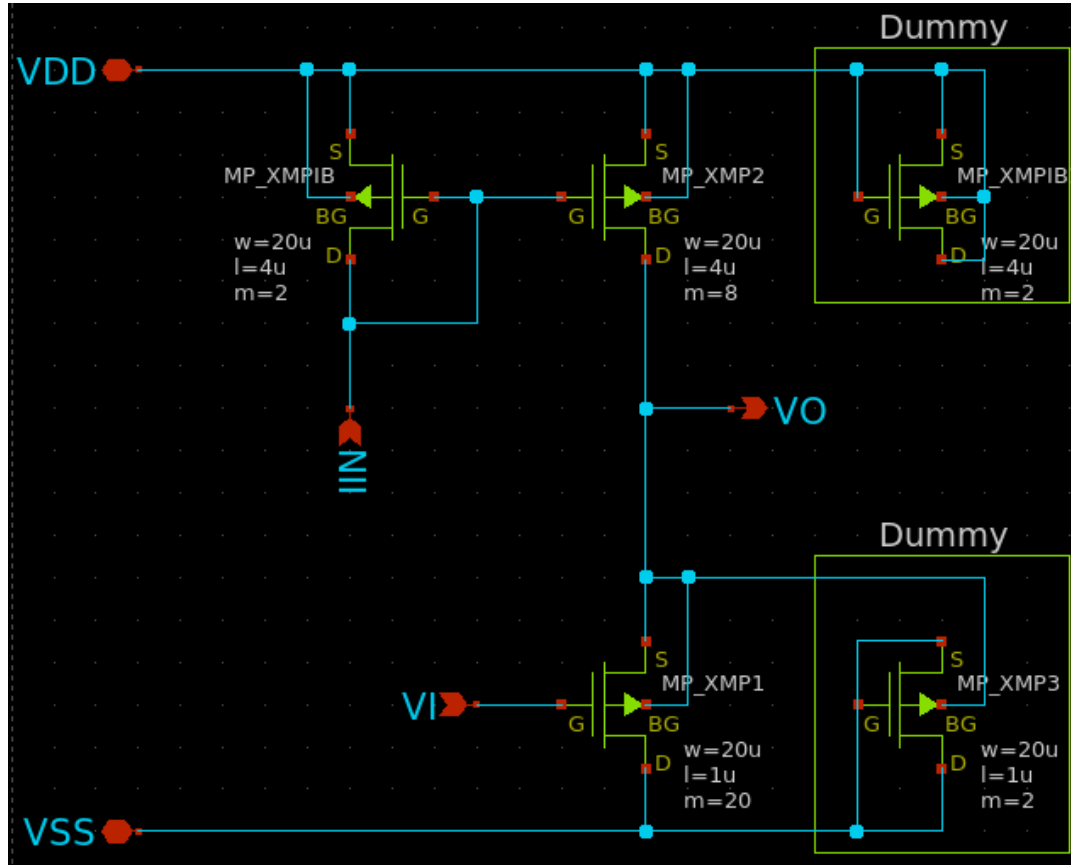


- Layout



psf (PMOS Source Follower)

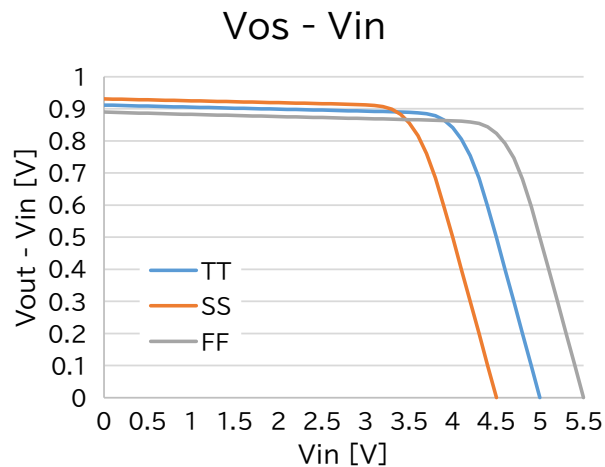
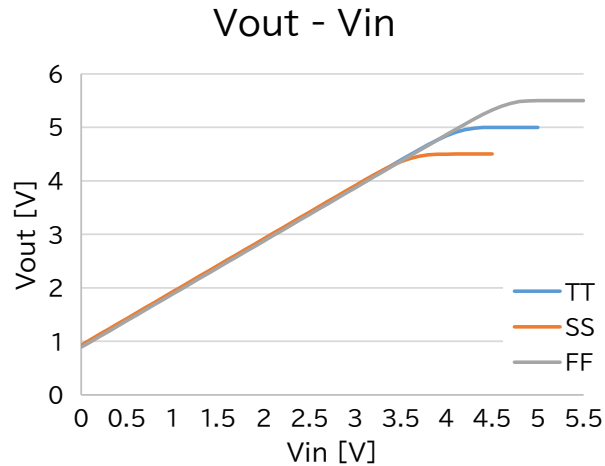
- Schematic (incl. layout-placed dummies)
- Layout



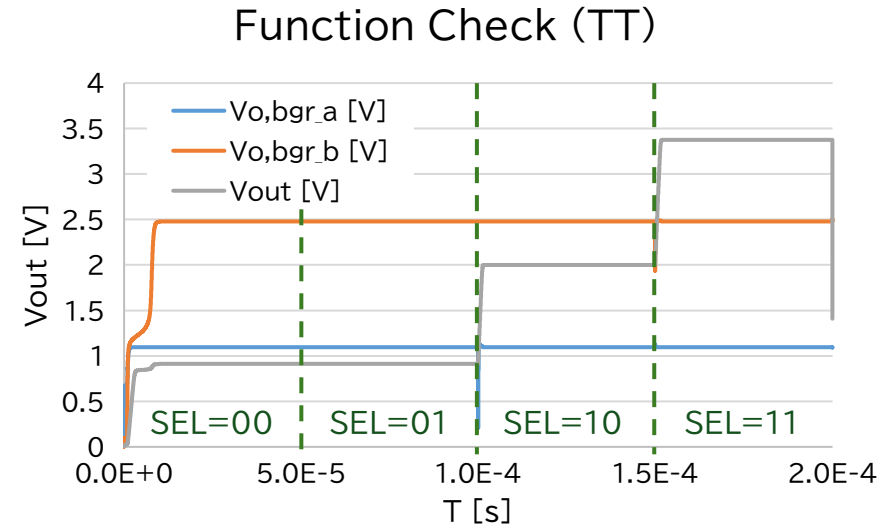
psf (PMOS Source Follower)

Teshbench: tb_psf.sch, tb_bgr_top.sch

- DC Analysis (w/bgr_b, ibuf)



- Tran. Analysis (bgr_top)



- SEL=0* $V_{out} = V_{ss} + V_{os,psf}$
- SEL=10 $V_{out} = V_{o,bgr_a} + V_{os,psf}$
- SEL=11 $V_{out} = V_{o,bgr_b} + V_{os,psf}$